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Facultad de Ingeniería Automática y Biomédica

Diploma Thesis

Automation of the subprocesses associated with line 320 of the 300 rolling mill at Antillana de Acero using the Siemens S7-1500 series

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Declaration of Authorship

I hereby declare that I am the sole author of this thesis and I authorize the Universidad Tecnológica de La Habana "José Antonio Echeverría" to make use of it as it deems appropriate, once it has been discussed and approved by the corresponding committee.

Dedication

To my mother, for her endless love and for always believing in me.
To my father, for being present in the most difficult moments and backing every decision I made.
To my grandmother, for her affection and her unconditional support.
To all my family and loved ones.

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To my mother, for giving me life and for accompanying me along this long path of my education, and for giving me the opportunity to become an engineer.

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To the workers of Antillana de Acero, and especially to the technician Yosbel, for teaching me everything about the rolling mill.

To all of them, thank you very much!

Abstract

This diploma thesis proposes the automation of subprocesses associated with line 320 of the 300 rolling mill at Antillana de Acero, using the Siemens S7-1500 series. The main objective is to modernize the currently obsolete automatic control system to improve efficiency and reduce production downtime. Line 320 operates with two GE Fanuc series 90-30 PLCs, which are outdated and face communication issues, making maintenance and operation difficult. The proposal includes migrating to Siemens S7-1500 PLCs and implementing two screens in a SCADA system to monitor and control the rolling process.

The thesis is structured into four chapters. In Chapter 1, the state of the art in rolling technology is described, both internationally and nationally, highlighting advancements in the steel industry, such as cold rolling and the integration of Industry 4.0 technologies. The rolling process at Antillana de Acero is also detailed, including the main components and instruments used. In Chapter 2, the automation proposal is presented, with the selection of the Siemens S7-1500 PLC and the design of flowcharts to control subprocesses in line 320, such as drags and shears. Chapter 3 addresses the monitoring interface and communication network, using Siemens WinCC V7.5 software for real-time supervision. Finally, in Chapter 4, a technical-economic analysis is conducted, evaluating the project's feasibility and its impact on production, the environment, and the company's economy.

The project concludes that modernizing the automatic control system will significantly improve the efficiency of the 300 rolling mill, reducing downtime and increasing product quality. Additionally, it highlights that the implementation will resolve current communication and maintenance issues, enabling Antillana de Acero to compete in more demanding markets.

Keywords: automation, rolling mill, PLC, Siemens S7-1500, SCADA, Antillana de Acero.

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Introduction

Rolling is an essential technique in the metallurgical industry, used to transform metallic materials into sheets or profiles with specific properties. This method, which dates back centuries, has evolved significantly with technological progress, enabling the production of materials with high precision and quality.

The "José Martí" Steel Company, known as Antillana de Acero, belongs to the GESIME Business Group under the Ministry of the Steel Industry (MINDUS). Its corporate purpose is the production of carbon steel billets, corrugated bars, and hot-rolled profiles to meet the demand of the national economy and for export.

It is located at Calle 20 #10522, Cotorro Municipality, Havana Province, and was founded on May 4, 1959, the year in which it made its first steel cast, opening a new era for steelmaking in Cuba. This backs more than fifty years of production experience and an active participation in both the domestic and foreign markets, mainly in the Caribbean, Central America, Europe, and Asia. Today the company is immersed in a modernization project and is the largest steel producer in the country. It produces and wholesales corrugated bars, plain bars, rolled profiles, and wire rod for the domestic market, and exports carbon steel billets through the Acinox Comercial company.

Mission of the organization

With more than 40 years of experience in the sector, the "José Martí" Steel Company (Antillana de Acero) produces carbon steel billets and hot-rolled profiles to satisfy the needs of our domestic and foreign customers and the country's commitments.

Vision of the organization

Antillana de Acero guarantees production levels of carbon steel billets of various grades and hot-rolled products, which enables it to achieve leadership in the Caribbean market with an important share of the Central American market. Its productions are backed by ISO 9001-2008, economic efficiency, reliability, high ethics in honoring our commitments, and competitive prices. Our workers, with a strong sense of belonging, are committed to our company's objectives and are a key factor in its competitiveness and productivity.

Problem situation

The finishing line L320 of the 300 rolling mill has two obsolete, discontinued GE Fanuc series 90-30 PLCs whose manufacturer no longer exists. These are modular PLCs using the SNP-X communication protocol, with which communication problems exist. Replacing and repairing the modules of the existing PLCs is difficult because they are obsolete. Technology that is too simple is used for a process that should have more redundancy and sophistication. The control is unsatisfactory and does not serve the process efficiently, causing stoppages, delays in fault repair, and an increase in the frequency of faults.

Problem to be solved

A migration to Siemens S7-1500 PLCs must be carried out as a modernization of the existing PLCs in the finishing and roughing lines of the 300 rolling mill. The implementation of a SCADA system is also considered in order to properly attend to the process.

General objective

To propose a modernization of the automatic control system for the subprocesses associated with line 320 of the 300 rolling mill at Antillana de Acero.

Specific objectives

- To research the state of the art.
- To describe the rolling process in detail, naming each component.
- To carry out the instrumentation survey as well as the modernization proposals.
- To propose the PLC programming through the flowchart of the algorithm.
- To develop the addition of two screens to a SCADA system.
- To carry out the technical and economic analysis.

Subject

Hot rolling process of steel bars for reinforcing bar (rebar).

Scope

The automation system of the subprocesses associated with line 320 of the 300 rolling mill at Antillana de Acero.

Hypothesis

Replacing the existing automation system with a more modern one in the subprocesses associated with line 320 of the 300 rolling mill at Antillana de Acero will improve the mill's production, thanks to a more robust and efficient system.

Methods used

Theoretical methods:

- Analysis and synthesis: the topic is broken down into parts and the information is integrated to present a complete view of the state of the art.
- Hypothetic-deductive: hypotheses are posed about the benefits of technological modernization and conclusions are deduced from success cases and technological advances.

Empirical methods:

- Expert criterion: the information is supported by references to the company's specialists who are present in and interact with the process.
- Simulation: simulation in WinCC V7.5 will be used for the programming and design of the SCADA.

Thesis structure

- Chapter 1: the state of the art, the description of the process, and its instrumentation survey will be presented.
- Chapter 2: the modernization proposal and the algorithm to be followed by the PLC, expressed in a flowchart.
- Chapter 3: the SCADA supervision system, its programming and screens, and the proposed communication system.
- Chapter 4: the economic analysis covering the cost of the modernization.

Chapter 1: Automation system of line 320 of the 300 rolling mill

1.1 Introduction

In the field of the metallurgical industry and materials processing, rolling mills play a fundamental role in the production of sheets, plates, and metal profiles. They are machines that shape steel through a plastic deformation process, obtaining products with specific dimensions and properties. The rolling process requires automatic control. Automation not only ensures greater precision and repeatability in the process, but also optimizes resource use, reduces production times, and minimizes human errors. In an increasingly competitive industrial environment, automatic control has become an indispensable tool for maintaining high standards of quality and productivity that are essential to the economy and national development.

In this chapter, an analysis of the state of the art of rolling mills will be carried out, exploring the most recent technological advances and trends in the design and operation of this equipment. In addition, the rolling process will be explored in depth, breaking down each stage. The instruments and control systems used will be detailed, and a proposal integrating advanced technological solutions for the modernization of the plant will be presented. Through this analysis, we seek to identify improvement opportunities that allow its operation to be optimized in an ever more demanding industrial context.

1.2 State of the art

1.2.1 International panorama

Rolling is a key process in steel transformation and the manufacture of metallic products. It is based on the deformation of the metal by the compressive force exerted by two rollers rotating in opposite directions which, through friction, change the dimensions of the steel being processed. Historically, and generally, this process has been carried out exclusively hot, since the material being transformed requires great flexibility and low residual stress. However, innovation has recently appeared in cold rolling.

The cold rolling sector has established itself as a fundamental pillar within the global steel industry, enabling the creation of steel products of excellent quality that are indispensable in multiple sectors. From the automotive industry to construction and electronics, this process guarantees precision, durability, and impeccable finishes, making it a key element of contemporary manufacturing.

During cold rolling, steel is subjected to the action of rollers at temperatures below its recrystallization point. This method not only strengthens the metal, increasing its tensile strength, but also improves its granular structure. As a result, a material with a superior surface finish and more precise dimensional tolerances is obtained compared with hot-rolled steel. Among its advantages are a more uniform surface and constant thickness, which are essential in industries where aesthetics and accuracy are priorities. Likewise, this type of steel offers greater strength and longevity, making it ideal for applications that must withstand extreme stresses and wear conditions. In addition, the process allows thinner and lighter sheets to be manufactured, which are in high demand in sectors such as vehicle body manufacturing and lightweight structures.

The adoption of Industry 4.0 technologies, such as the Internet of Things (IoT), artificial intelligence, and big data analytics, is revolutionizing this market. These tools enable greater automation, more precise process control, and better-founded decisions. For example, IoT sensors can monitor critical variables in real time, such as temperature, pressure, and roller

alignment, generating data that allow rolling to be optimized. For their part, artificial intelligence and machine-learning algorithms analyze this information to anticipate equipment failures, improve maintenance plans, and increase operational efficiency.

Numerous companies worldwide have already adopted these innovations to expand their market presence and offer cutting-edge solutions. A notable example is ArcelorMittal, one of the world's leading steel producers, which uses cold rolling to manufacture high-end materials. Another reference is Nippon Steel Corporation, based in Japan, recognized as a leader in cold-rolled steel production. Similarly, Tata Steel, based in India, stands out for its technology

in this field, supplying steel for a wide variety of applications, from the automotive industry to construction.

Cold rolling has not been the only revolutionary innovation in recent years: the dry-film laminator sector is recording notable growth, driven by technological advances in the rolling field and their application in various industries. These machines play a crucial role in sectors such as electronics, automotive, packaging, and healthcare, where they are used to apply protective coatings onto surfaces with high precision and efficiency. Unlike wet rolling methods, dry-film laminators offer benefits such as faster production cycles, less material waste, and more durable end products. Globally, this market is expanding as manufacturers seek economical solutions that improve both the appearance and the performance of their products.

To support the above, leading companies that have adopted this technology can be mentioned. For example, the KETE Group, specialized in the packaging industry, provides laminating machines capable of working with material combinations such as plastic-plastic, paper-plastic, and aluminum-plastic. DirectIndustry, for its part, is a prominent supplier of dry rolling equipment designed for plastic, paper, and aluminum foil applications. Finally, Shanghai Outex stands out for its manufacture of high-quality dry laminating machines adaptable to a wide range of industrial needs.

1.2.2 National environment

Since 2019, the company Acinox-Las Tunas has been working on an innovative project that seeks to integrate science and technology to drive development and improve the efficiency of one of Cuba's most important metallurgy plants. This initiative, linked to the National Development Program "Automation, Robotics and Artificial Intelligence", uses Advanced Automation in steel elaboration and refining processes. Thanks to research carried out within master's degree programs, conceptual solutions have been proposed that address various production challenges.

At the Antillana plant, the drive and automation system of line 400 of the 300 rolling mill was modernized, which allowed restoring the production of half-inch steel profiles for both the domestic market and export. For this renewal, ABB DCS800 digital DC converters, a Siemens Simatic S7-300 CPU 315-2 PN/DP programmable controller, and a monitoring system based on WinCC version 7.0 + SP2 software were implemented. In addition, a supervision system was developed for the reheating furnace using Schneider Electric's Citect SCADA. Motivated by these successful results, the company plans to extend this modernization to other areas of its operations.

1.3 Previous work

Erik García Vázquez (2011), in his thesis, developed the design of the speed control system for the motors of the finishing line of the rolling train of the 300 rolling mill at Antillana de Acero.

Jeffrey Ugarte Vargas (2019), in his thesis, made a proposal for the modernization of the automation system associated with the subprocesses of line 320 of the 300 rolling mill at Antillana de Acero.

Viana Menezes Demba (2020), her thesis had as its main objective the design of the automation and electrical drive system in the head-and-tail cutting area of line 320 of the 300

rolling mill at Antillana de Acero.

1.4 Process description

The steel billet leaves the reheating furnace at a temperature of around 1200 °C and is carried by the roller tables to the roughing stand, through which it passes 6 times with the help of the rocking table — a process needed to stretch it. It then follows the roller table to the bump, where the roller table is rotated in the opposite direction by manual drive. It next meets a diverting switch, like the one used on railway lines; on the return path it passes through the roughing line, where it crosses 4 stands. Stands 2 and 3 are coupled to the same motor, whereas stands 4 and 5 each have an independent motor. To continue the rolling process, the billet must be incorporated into line 320, which is parallel to the path already covered; to achieve this, the billet enters the bending shoe — no more than a curved metal piece that makes the billet create a curvature as it passes through and enter the drag TR2, continuing along the line. At the end of the bending shoe there is a photocell that, upon seeing the tip of the billet, gives a signal to the bending shoe; it then lifts and lets the billet stretch to prevent it from breaking due to excess tension. The roller table of line 320 moves the billet to the aforementioned drag TR2, which brings it into contact with the shear CV2, which cuts the head and tail of the billet. Next there is a tray that, actuated by a solenoid, opens to let the tail and head out and closes once the billet has passed. It then enters stands 6 to 11, with stand 11 being the only vertical stand used in the process. After the last stand there is a photocell and the drag TR3, which serve the cutting cycle performed by the shear CV3, giving the billet the longest possible length for the cooling bed. Once finished, it passes to the ejector, which has its own shaft control that throws the billet according to its thickness onto the cooling bed, which rotates the final product and moves it towards a parallel platform where the cold shear cuts the final product according to customer requirements.

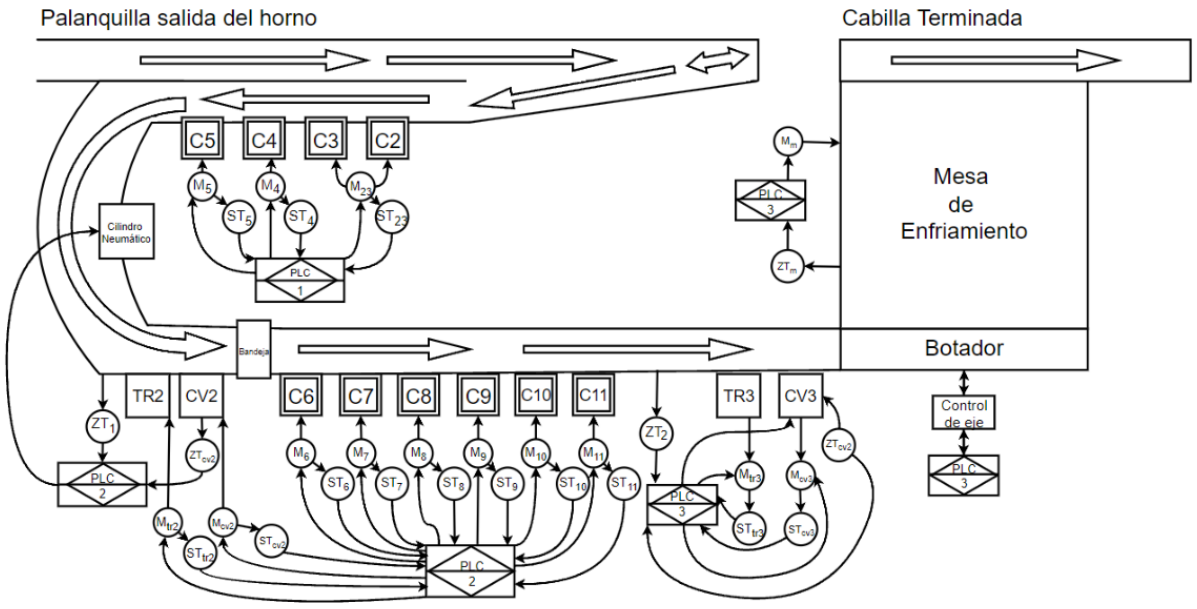


Figure 1.1 Functional diagram (field view).

1.5 Instrumentation survey

The 6 motors of line 320 (or finishing line) are of the Elektro Maschinenbau brand. They are direct-current, separately excited machines. Their nameplate data are: speed 320 to 960 RPM, excitation current up to 23 A, nominal power 630 kW, nominal current 1825 A, and nominal voltage 600 V (see Annex 1).



Figure 1.2 Elektro Maschinenbau DC motor.

ABB DCS800 converters are used for the stand motors.



Figure 1.3 ABB DCS800

Four Gefran TPD32-EV converters are used for the drags, the shear CV2, and the ejector (see Annex 2).



Figure 1.4 Gefran TPD32-EV

Two ABB ASC800 frequency converters are used to drive the roller table of the cooling bed and the shear CV3.



Figure 1.5 ABB ACS800

A Dinamo Tachimetrica tachometer is used as a transmitting sensor, giving a signal of 0.06 V/ RPM.



Figure 1.6 Dinamo Tachimetrica.

In addition, to display the speeds on the console panel an ELCIS 1/X115T-1000-1230-BZ-C-CL-

R encoder is used.



Figure 1.7 ELCIS 1/X115T-1000-1230-BZ-C-CL-R encoder.

The speed ratio between one motor and another is controlled by means of BMB 4.082.0 A4 proportional-integral (PI) analog regulators.

It has four ID 2000 series photocells from the Danieli brand, designed to detect the presence of bodies emitting hot infrared radiation.

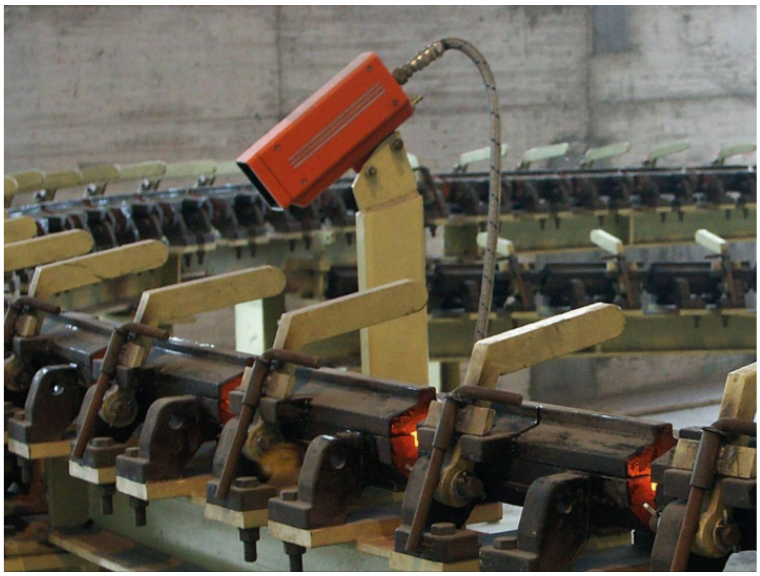


Figure 1.8 Danieli photocell.

Five Siemens 3RG4 013-3JB00 inductive sensors are used on each shear, one pair on the cooling bed, and one on the ejector shaft control, with a 5 mm sensing distance, M18 brass nickel-plated cylinder, 15-34 V DC, 2 wires, 1 N.O. (normally open) contact, 25 mA load current, 400 Hz switching frequency, and a 12 mm plug connector.



Figure 1.9 Siemens inductive sensor.

The two aforementioned GE Fanuc 90-30 programmable controllers have a total of 3 high-speed counting modules, 14 digital input modules, 2 analog output modules, and 13 digital output modules.



Figure 1.10 GE Fanuc 90-30 PLC

1.6 Some standards used

- IEC/TR 62442-3-1: this standard provides a detailed assessment of cybersecurity technologies and tools, offering valuable guidance for manufacturers, suppliers, and security professionals in protecting IACS (industrial automation and control systems) against cyberattacks, with the aim of preserving the integrity of industrial systems and minimizing operational and safety risks.

- IEC 62381: the annexes of this standard include forms that can be used in test procedures. Purchasers of this standard may reproduce these forms for their own purposes, always in the necessary quantity. The purpose of this standard is to optimize and streamline the negotiation phase and to facilitate a clear and mutual understanding of the scope of the activities corresponding to each party.

- IEC 62337: this standard seeks to improve and streamline the negotiation phase, fostering a clear and mutual understanding of the scope of the activities and responsibilities of each party involved. In the process industry there is a growing trend to assign the complete construction of plants to contractors under global modalities, such as turnkey or similar. However, experience has shown that both the process industry and the contractor typically face prolonged and costly

discussions to define precisely and without ambiguity the tasks corresponding to each one, as well as their responsibilities for ensuring the successful delivery of the plant.

1.7 Partial conclusions

This chapter analyzed the technological progress of rolling processes in the global and national contexts, identifying both the achievements reached and the persistent limitations in the development of automated and intelligent systems (SAI). Through a methodological approach, the inherent complexity of the process was addressed by means of a systemic analysis, complemented by the design of a functional diagram that facilitates its structural understanding. Additionally, a diagnostic evaluation of the current state of the instrumentation in the rolling mill was carried out, evidencing technological gaps and optimization opportunities. These results underline the need to integrate innovative solutions in order to overcome operational challenges and consolidate the transition towards more efficient and adaptive systems in the metallurgical industry.

Chapter 2: Automation proposal

2.1 Introduction

In the context of industrial automation systems, the programming of Programmable Logic Controllers (PLC) plays a crucial role in ensuring the efficiency, safety, and reliability of processes. This chapter presents the design and development of a flowchart for the programming of a PLC that controls a rolling line, specifically focused on the operation of the auxiliary elements of the process, since they enable the transport and precise cutting of the material, ensuring a continuous and synchronized flow in the production line.

The proposed flowchart integrates the essential functionalities of start, stop, and alarm management, considering the interactions between the drags and the shears, as well as the operating conditions and the possible critical events that may arise during the process. Key aspects such as the logical operating sequence, fault detection, and the system's response to abnormal situations are also addressed. This approach not only seeks to optimize the performance of the rolling line, but also to ensure operational safety and facilitate preventive maintenance.

Through this chapter, a detailed guide is provided for the implementation of the program in the PLC, backed by a technical analysis that justifies each design decision. The objective is to contribute to the progress of automation in the rolling industry, offering a robust and scalable solution that can adapt to different configurations and production requirements.

2.2 PLC selection

To replace the 2 GE Fanuc PLCs, the CPU 1516-3 PN/DP from the SIMATIC S7-1500 line was selected for its characteristics, among which are: internal program memory of 1.5 MB and data memory of 5 MB; PROFINET (PN) and PROFIBUS (DP) communication interfaces; and I/O modules, supporting up to 32 modules. This CPU will be installed in the central machine control room with a remote station to control all the modules that will be used, while in the yellow cabin — where the GE Fanuc CPU is currently located — another remote station will be installed. This will represent a saving in the installation of PROFINET and PROFIBUS cables, since the CPU supports both communications: it will be closer to the drive, with which it will communicate via PROFIBUS, while it will communicate with the remote station via PROFINET.



Figure 2.1 CPU 1516-3 PN/DP

Sixteen DI 16x24VDC digital input modules will be used; each includes 16 digital inputs. It

requires a voltage of 15–30 VDC to recognize a logic 1 state and 5 VDC for state 0. The typical input current in state 1 is "d 2–15 mA (at 24 VDC). For digital outputs, sixteen DQ 16x24VDC/0.5A modules are proposed; each has 16 digital outputs. It delivers 24 VDC (range: 20.4–28.8 VDC) in state 1 and "d 2 VDC in state 0, and the maximum current per channel is 0.5 A in the active state.

For the high-speed counting modules, seven TM PosInput 2 digital input modules will be used. Each module has two independent channels; its signal type is compatible with incremental encoders (A, B, R) and NAMUR signals. The maximum frequency is 1 MHz (for TTL/HTL signals), and it features bidirectional counting, interpolation, value comparison, and integrated diagnostics.

In the remote station in the yellow cabin as well as in the one located right below the CPU, an IM 155-6 PN ST communication interface (see Annex 3) with the PROFINET protocol will be used to communicate with the CPU; it is worth noting that it supports up to 32 modules.

For the case of the remote station in the yellow cabin only, a PS 60W 24VDC power supply with 120/230 VAC input and 24 VDC / 2.5 A output will also be needed, suitable for powering 32-channel modules. For the digital input modules, the use of three DI 32x24VDC BA with 32 input channels (24 VDC) and short-circuit and open-wire detection is proposed. For the digital output modules, four DQ 32x24VDC/0.5A BA are required, with 32 output channels (24 VDC, 0.5 A per channel) and short-circuit and overload protection. All modules belong to the ET 200SP line, optimized for distributed systems.

	Cantidad	Modelo/Referencia	Descripción
	1	6ES7516-3AP03-0AB0	Para el cuarto central de control de máquinas, soporta hasta 32 módulos.
	1	6ES7954-8LC03-0AA0	Tarjeta de memoria para S7-1x 00 CPU/SINAMICS, 4 MB.
	16	6ES7521-1BL00-0AB0	Módulo de entrada digital de 16 canales (24 VDC, detección de 15–30 VDC para "1").
	16	6ES7522-1BH00-0AB0	Módulo de salida digital de 16 canales (24 VDC, 0.5 A por canal).
	7	6ES7550-1AA00-0AB0	Módulo de conteo rápido con 2 canales (1 MHz, compatible TTL/HTL/NAMUR).
	2	6ES7155-6AU00-0CN0	Interfaz de comunicación PROFINET para estac (soporta 32 módulos).
	1	6ES7505-0RA00-0AB0	Fuente de alimentación para backplane (24 VDC / 2.5 A).
	3	6ES7521-1BL10-0AA0	Módulo de entrada digital de 32 canales (24 VDC, detección de cortocircuitos).
	4	6ES7522-1BL10-0AA0	Módulo de salida digital de 32 canales (24 VDC, 0.5 A por canal).
e	39	6ES7592-1AM00-0XB0	Para módulos DI/DQ 16x24VDC, DI/DQ 32x24VDC BA y TM PosInput 2
	3	6ES7590-1AB60-0AA0	2 para rack central (CPU) y 1 para estación rem (IM 155-6 PN ST).
	1	SCALANCE XB005	Conmutador de 5 puertos para red PROFINET e CPU e IM 155-6 PN ST.
	10	6GK1901-1BB10-2AA0	Conector Profinet RJ45 plug 180 2x 2, conector enchufable RJ45 (10/100 Mbit/s)
	15	6ES7972-0BA52-0XA0	Conector Profibus in enchufe PG, salida de cab ángulo de 90 grados, hasta 12 Mbps.
	1		Software para configurar el sistema de monito
	1	STEP 7 Professional	Software para configurar CPU S7-1500 y módul ET 200SP.
	100 mt	-	Para conexion entre la CPU y la estación remo
	60 mt	-	Para conexión entre los actuadores y la estaci
	4	TPD32-PB	Trajeta de comunicacion PROFIBUS para los convertidores Gefran TPD32-EV
-02	6	RDCO-02	Trajeta de comunicacion PROFIBUS para los convertidores ABB DCS800
-02	2	NNPB-02	Trajeta de comunicacion PROFIBUS para los variadores de frecuencia ABB ACS800

Table 2.1 Required modules and accessories.

2.3 Flowcharts

The first flowchart designed represents the logical sequence that the PLC must follow for the start-up of the rolling mill. This sequence is guided by the start signal given by the operator in the yellow cabin, and additionally requires a signal from the furnace indicating that a billet with the appropriate temperature level for rolling will be sent along the roller table. Once these requirements are met, all the stand motors as well as the roller table are initialized, and their direction of rotation is set so that the billet heads to the left (seen from the cabin).

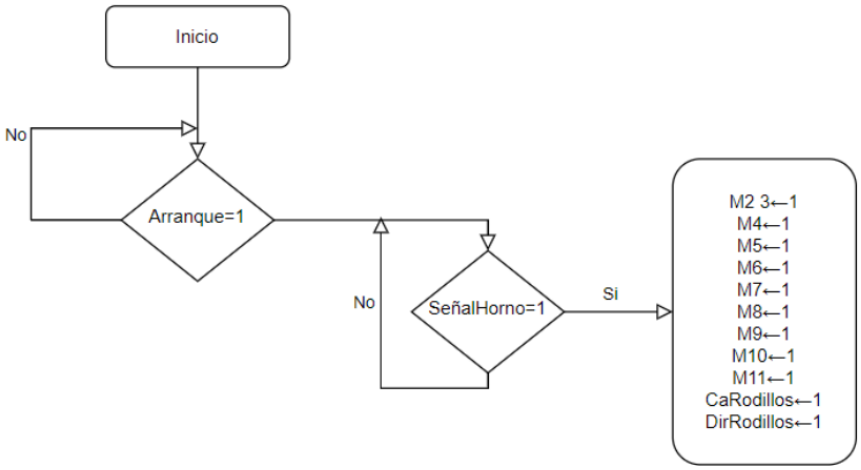


Figure 2.2 Start-up flowchart.

The first auxiliary element to be developed is the drag TR2, located just after the repeater that curves the billet. This drag will require the rising edge of the photocell ZT1; a timer previously calculated as a function of the roller table speed is activated and, when it finishes, an output signal is issued that enables the solenoid. This makes the drag rolls close, trapping the billet. Next, another timer is enabled that waits for a time that depends on the speed and has to do with the length between the drag and stand 6, such that it guarantees that the stand has grabbed the billet. At this point the drag opens, fulfilling its function, and waits for the next billet.

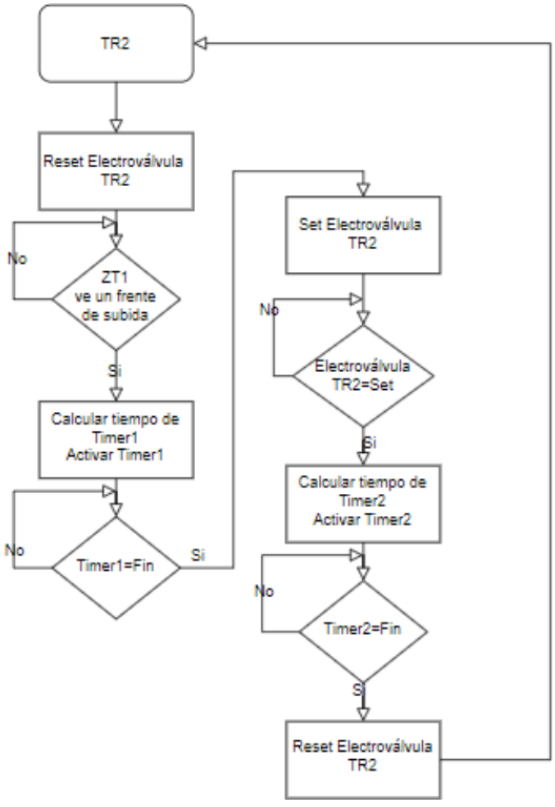


Figure 2.3 Flowchart for the drag TR2.

The shear CV2 is in charge of cutting the tail and head of the billet, since these zones cool down more rapidly and could damage the equipment if they are not removed. According to the signal of the first photocell, a time is calculated that includes the time the billet takes to travel from the photocell to the shear plus the time corresponding to the length of the head to be cut. Knowing the constant value of the roller table speed, this calculation is very simple in the case that automatic cutting is required. For manual cutting, it will only be necessary to enter the time

value considered by the operator. When the time elapses, the shear makes a cut and, using the induction sensor ZTcv2 — which sends a signal upon detecting the shear's cut — the solenoid valve is reset. For the tail cut, both the sensor ZTcv2 and ZT1 must see a falling edge indicating that the end of the billet is approaching. At this point the timer calculation is repeated for the automatic case, while the manual case waits for the operator's command. The shear's cut and the solenoid-valve reset are repeated upon seeing the signal of the sensor ZTcv2, and thus the head-and-tail cutting process ends, leaving CV2 ready for the next billet.

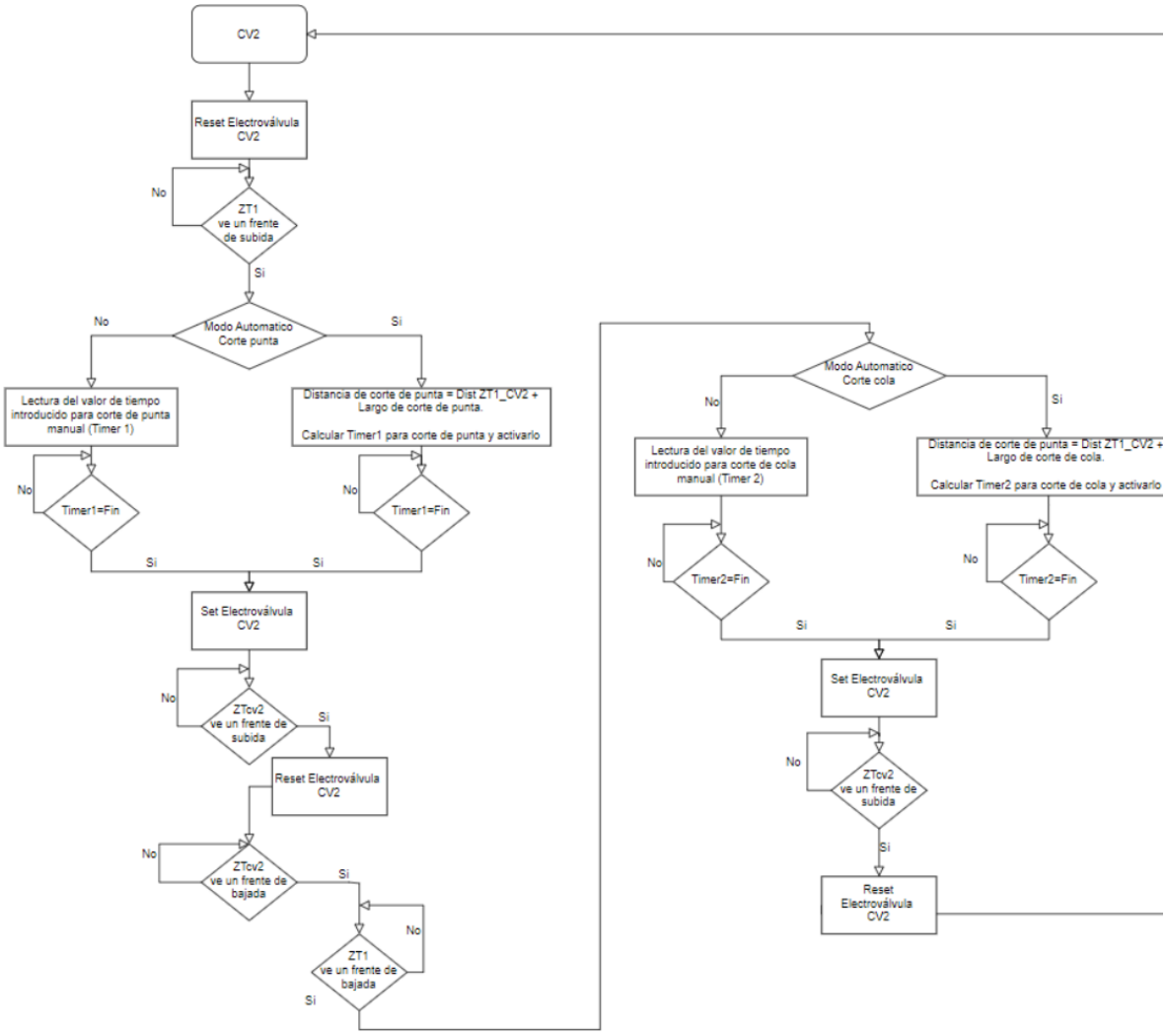


Figure 2.4 Flowchart for the shear CV2.

The drag TR3 follows a very similar algorithm to that of TR2, with the difference in the calculation of the times in which it grabs and releases the billet, since — unlike TR2 — this drag has the task of taking the billet to a shear that must make a cut to measure.

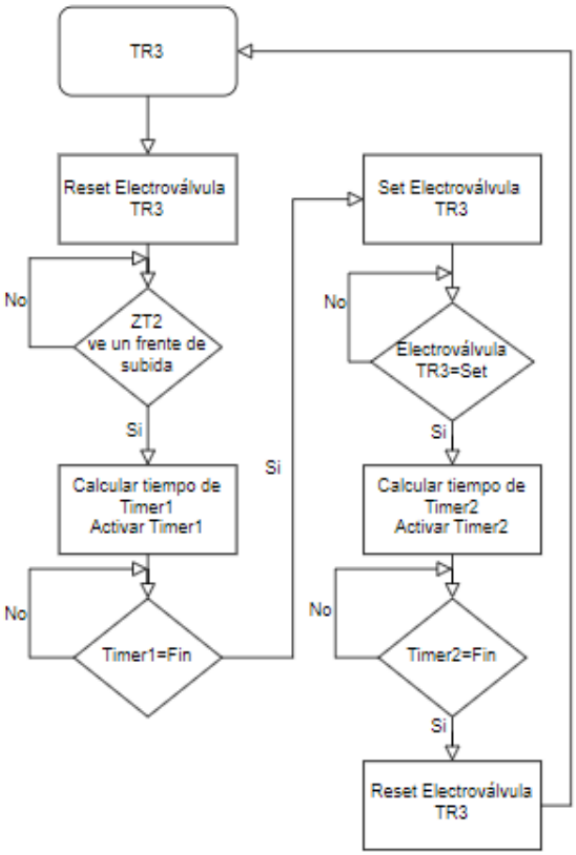


Figure 2.5 Flowchart for the drag TR3.

The shear CV3 makes a cut to measure on the hot rebar, which allows the profile to fit correctly inside the cooling bed. This algorithm includes a cut counter to keep track of how many rebars have been cut and are in the cooling process. From the rising edge of the second photocell ZT2, a timer is started with the time the billet takes to reach the shear. Then, depending on whether the photocell detects the billet or not, a cut to measure is made based on a time calculation that considers the desired length and the speed of the roller table. Should the photocell stop detecting the billet, it will be calculated whether the remaining profile is longer than the desired length. If it is not, the cut will not be performed and the algorithm will wait again for the positive pulse of the photocell, which indicates the arrival of the next billet. Otherwise, the cut will be carried out, and afterwards the signal of the photocell ZT2 will be waited for in the same way.

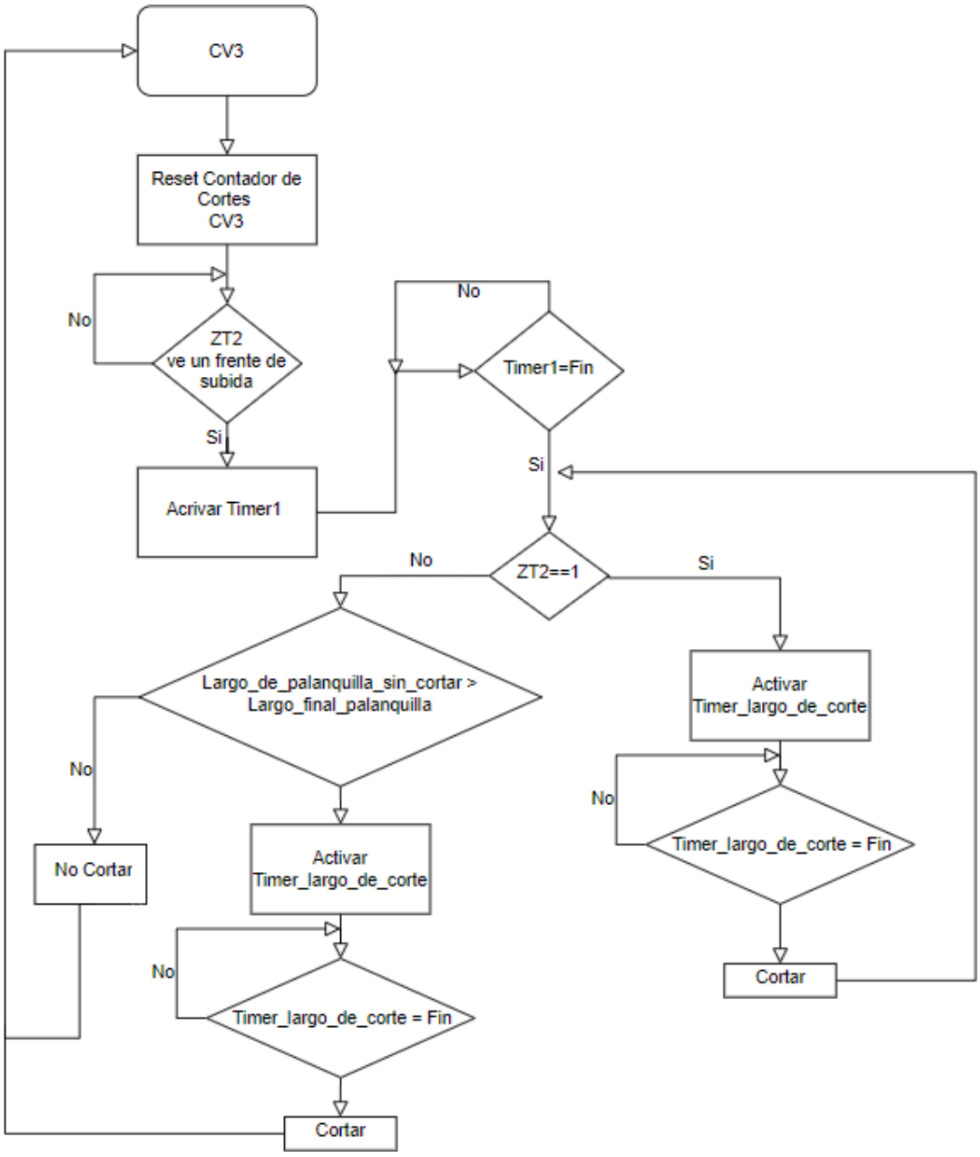


Figure 2.6 Flowchart for the shear CV3.

The following flowchart describes the stop process corresponding to the subprocesses before and after line 320, the stop conditions being either a confirmation by the operator or an alarm.

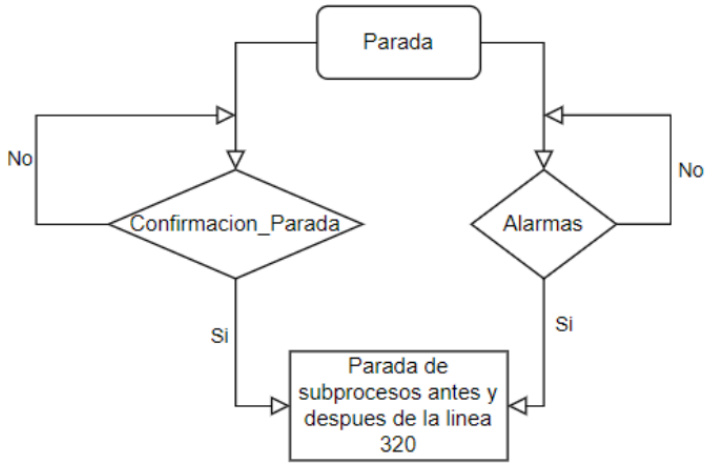


Figure 2.7 Flowchart for the stop of line 320.

Lastly, the two alarms corresponding to the subprocesses before and after line 320 are

described in these diagrams. These are the actuation of a breaker for motor voltage protection, and the signal from the flow switch installed on both drags, which gives a signal once the lubrication flow to them drops below the established standard levels.

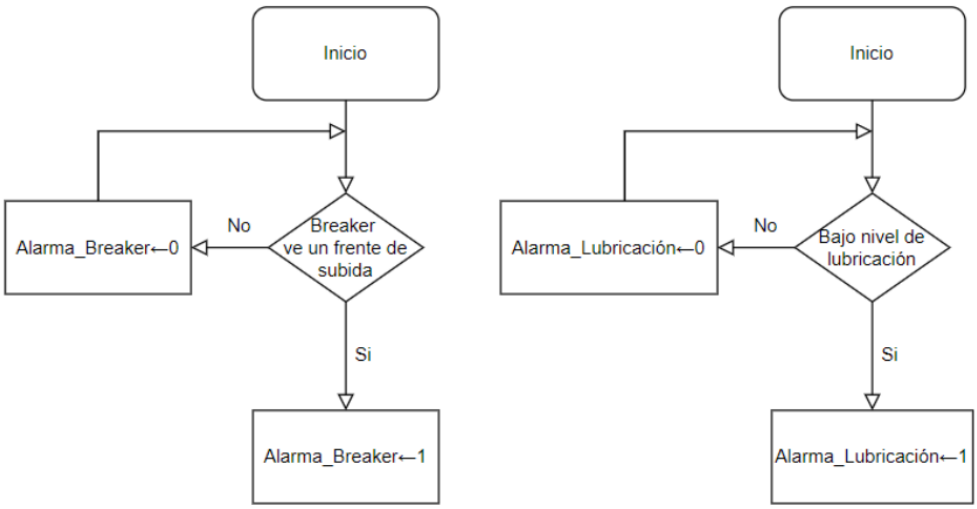


Figure 2.8 Flowchart for the alarms of line 320.

2.4 Partial conclusions

In this chapter, the design and control logic for the rolling and cutting systems has been presented, focusing on the operation of the shears CV2 and CV3 as well as on the synchronization of the drags and the roller table. Through detailed flowcharts and algorithms, a clear and efficient sequence has been established for the start-up, cutting, and handling of the billets, considering key aspects such as transport time, signal detection through photocells, and alarm management. The shear CV2 is responsible for removing the heads and tails of the billets, ensuring that the equipment is not damaged due to the accelerated cooling in these zones. For its part, the shear CV3 makes cut-to-measure operations to adapt the rebars to the dimensions required by the cooling bed. Both processes depend on precise calculations based on the speed of the roller table and on the detection of the billets through sensors, which guarantees synchronized and efficient operation. In addition, the importance of the auxiliary elements — such as photocells and induction sensors — has been highlighted: although they do not perform the main function, they are indispensable for the correct operation of the system. These components enable the automation of the process, reducing manual intervention and minimizing errors.

Chapter 3: Monitoring interface and communication network

3.1 Introduction

This chapter explores the features of WinCC software, developed by Siemens, which enables real-time visualization through customizable graphical interfaces, integrating data from PLCs and field devices via protocols such as PROFINET and PROFIBUS. Likewise, it includes advanced tools for alarm management, historical data collection, report generation, and access control, guaranteeing safe and efficient operations. In this context, WinCC V7.5 is used to supervise and control a 300 rolling mill, although it is stressed that the current system does not fully comply with the functionalities of a SCADA, mainly due to the absence of a real-time fault diagnosis and correction algorithm. In addition, the incorporation of new screens and monitoring variables to manage the auxiliary equipment located before and after line 320 is detailed below. A proposal to optimize the communication network through a client-server architecture, using PROFIBUS and PROFINET networks in order to increase the speed and flexibility of the connections, is also presented. Lastly, the concept of the automation pyramid is introduced, which represents the hierarchical levels of an industrial system, ranging from sensors and actuators at the field level to corporate management at the enterprise level.

3.2 Software features

WinCC, developed by Siemens, is a robust and flexible SCADA software used to create monitoring and control systems in industrial environments. It allows real-time visualization of processes through customizable graphical interfaces, integrating data from PLCs and field devices via protocols such as PROFINET, PROFIBUS, and OPC UA. WinCC offers advanced tools for alarm management, historical data acquisition, report generation, and access control, ensuring safe and efficient operations. Its scalability makes it ideal for applications in manufacturing, energy, water, oil & gas, and smart buildings, providing a reliable solution to optimize processes and improve industrial productivity. In this case we will use WinCC V7.5.

3.3 Screen description

In the center there is an existing monitoring system where the entire 300 rolling mill is attended with real-time values and is capable of controlling them. However, it cannot be considered a SCADA system due to the lack of a fault diagnosis and identification algorithm and of real-time correction actions. In this way it will be simply referenced as a monitoring system.

In the existing monitoring software there was no screen dedicated to the processes before and after line 320; therefore, as part of the objectives, monitoring variables and two screens were added for the control of the auxiliary equipment. These variables fulfill specific functions that are clarified in the flowcharts.

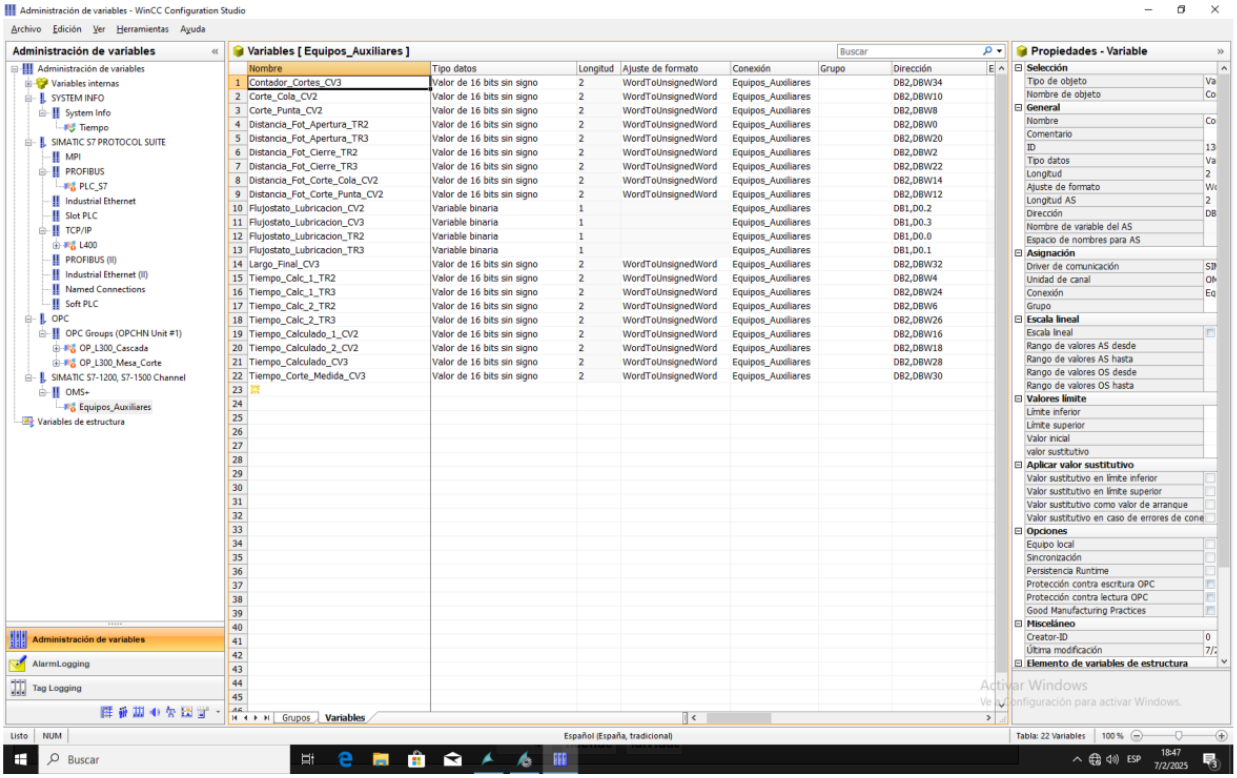


Figure 3.1 Added variables.

For the first screen, the control focuses on the zone before stand 6, namely the drag TR2 and the shear CV2, while the second added screen deals with the equipment after stand 11: the shear CV3 and the drag TR3. The motor parameters are displayed, in addition to process-specific variables such as the time of each cut, the distance in meters of those cuts, and timers. Also, on these screens the calibration error indicator is added, also configured as an alarm variable so that this decisive factor can be attended promptly. It is worth noting that these screens are accessed by pressing the TR2 and CV2 buttons for the first screen, and CV3 and TR3 for the second, from anywhere in the program.

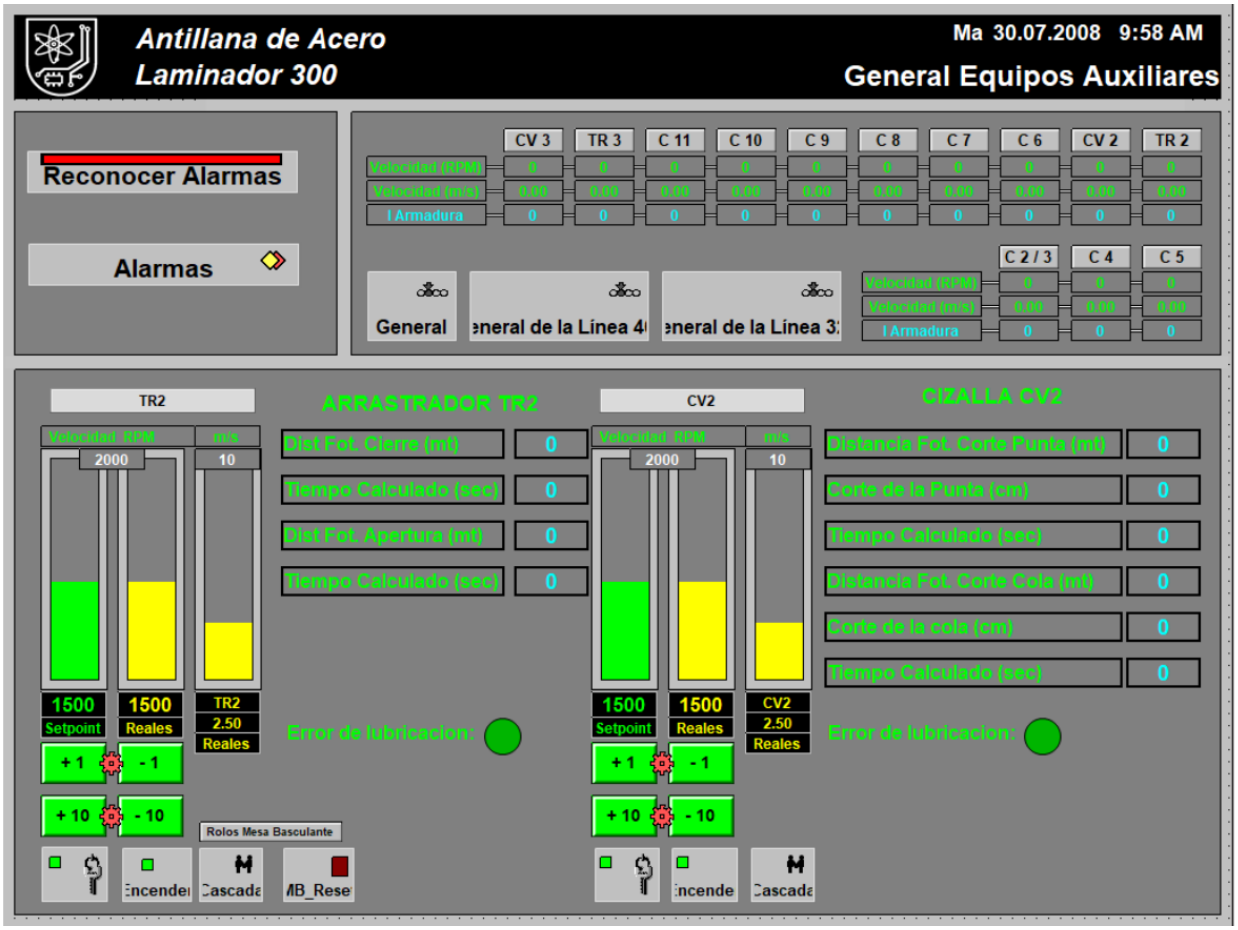


Figure 3.2 Screen for the auxiliary equipment before line 320.

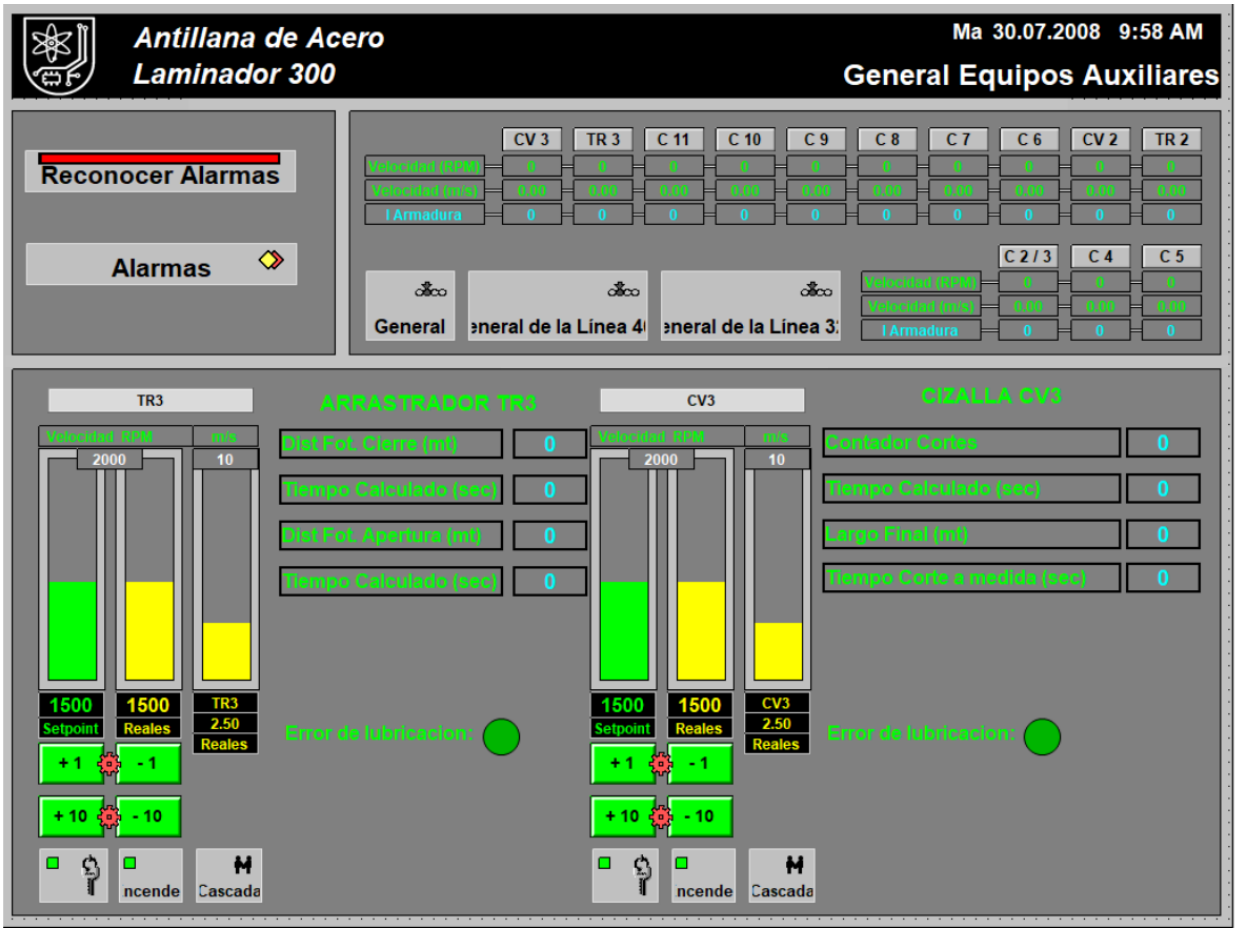


Figure 3.3 Screen for the auxiliary equipment after line 320.

3.4 Communication network

Communication on line 320 operates using the SNP-X protocol, which has limitations for data exchange. The lack of compatibility between the GE Fanuc PLCs and WinCC made it necessary to resort to the OPC standard as a solution to facilitate communication.

In the proposal, improving this communication was considered by establishing a client-server architecture. For this, it will be necessary to propose an HP EliteDesk 800 G6 Tower PC (see Annex 4), which is a robust and versatile equipment that offers excellent performance for server applications in controlled environments. It is equipped with 10th-generation Intel Core i5 processors, which guarantees high processing power to handle demanding tasks. Regarding RAM memory, it supports up to 64 GB DDR4, allowing efficient handling of data and real-time applications. In terms of storage, it has a configurable SSD that can range from 256 GB to 1 TB, providing speed and capacity for the operating system and critical applications. In addition, it offers the option to add a 1 TB or larger HDD, ideal for historical data storage and backups. The integrated network card is Gigabit Ethernet, ensuring a fast and reliable connection for network communication. The equipment is compatible with operating systems such as Windows 10 Pro or Windows Server, which makes it suitable for enterprise and industrial environments. Regarding connectivity, it includes multiple USB ports (including USB-C), HDMI and DisplayPort, in addition to allowing the installation of additional cards for industrial protocols such as RS-232/485 or PROFINET. Its cooling design is efficient and it features quiet fans, which contributes to stable and prolonged operation. In terms of security, it incorporates TPM 2.0 for data encryption and advanced protection features, guaranteeing the integrity of the information. Although it is not specifically designed for extreme industrial environments, its robust construction makes it adequate to operate in controlled conditions, offering an ideal balance between performance, reliability, and cost.

The PCs and the PLCs will be connected by a PROFIBUS network to increase transmission speed and make the connections more versatile, while the PLCs will be connected to the different actuators by PROFIBUS because that is the technology these devices support. In this way, having the remote stations will represent a saving in modules and components and will be much more durable than the simple connection that existed before. Internally, communication between the modules and the PLC CPU is managed through the SIEMENS hardware's internal bus.

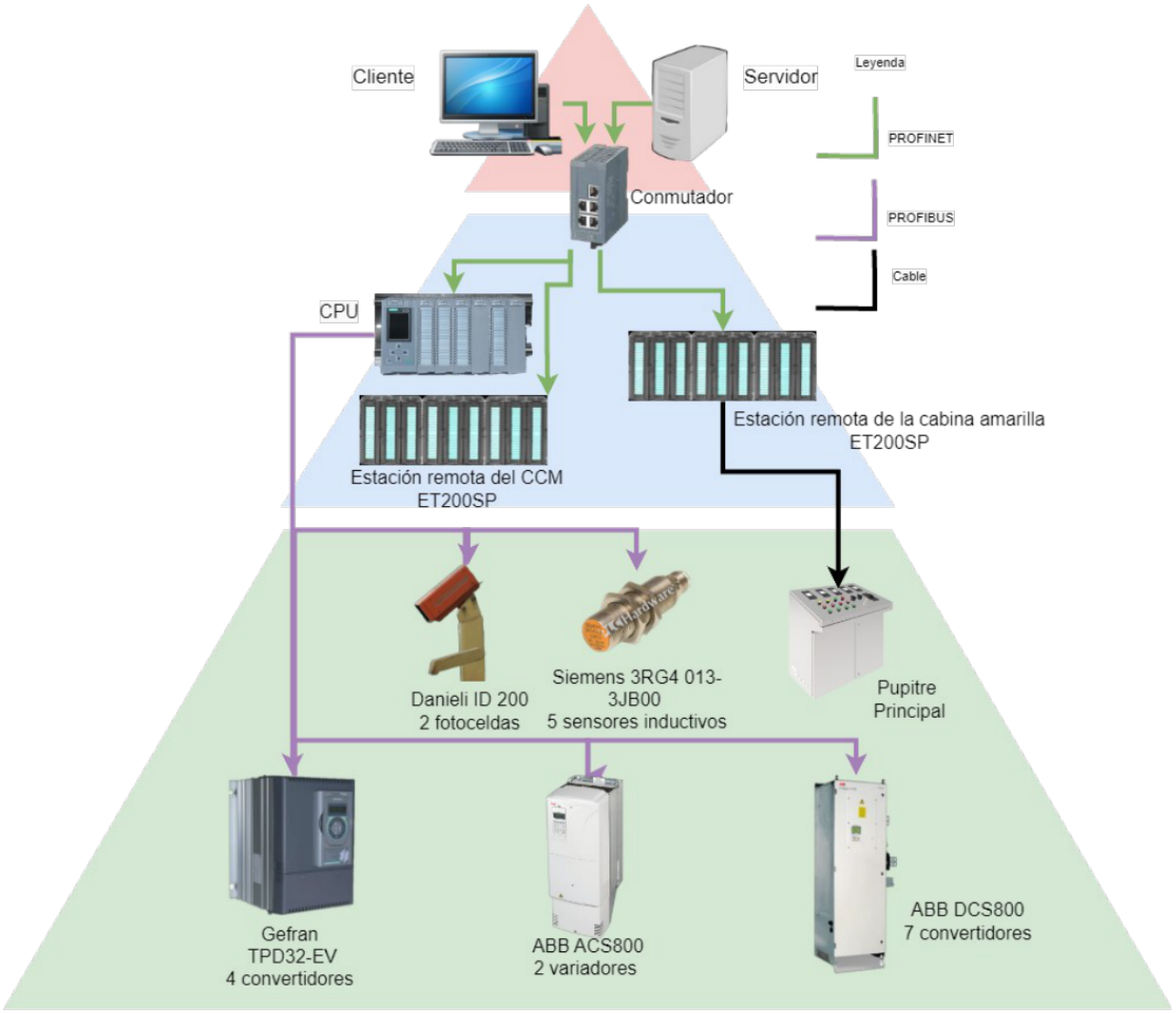


Figure 3.4 Automation pyramid.

PROFIBUS (Process Field Bus) is a fieldbus standard widely used in industrial automation. Based on RS-485 or optical fiber, it allows fast and reliable communication between devices such as PLCs, sensors, and actuators. It works in a master-slave topology and stands out for its robustness in harsh industrial environments. It is used in discrete manufacturing and continuous process applications. PROFINET is the Ethernet-based industrial communication protocol for Industry 4.0. It offers real-time communication (RT/IRT), integration with IT networks, and high speed (up to 1 Gbps). It supports plug-and-play configuration and advanced diagnostics, making it ideal for complex systems such as robotics or automated production lines. It is the modern evolution of PROFIBUS, adapted to the needs of digitalization.

The automation pyramid represents the hierarchical levels of an industrial system:

- Field: sensors/actuators.

- Control: PLCs.

- Supervision: SCADA, HMI, and monitoring.

- Planning: MES and production management.

- Enterprise: ERP and corporate management.

Each level communicates vertically, optimizing processes from plant operation to strategic decision-making. In the center, the last 2 levels are truncated; it would be important to propose their integration in future projects.

3.5 Partial conclusions

In conclusion, the use of WinCC software stands out as an essential tool for supervision and control in industrial environments, offering a scalable and reliable solution to optimize processes and improve productivity. Although the current monitoring system of the 300 rolling mill does not fulfill all the functions of a SCADA, new screens and monitoring variables were added, representing a significant advance in the improvement of the control of the auxiliary equipment before and after line 320. It was proposed to improve the communication network through a client-server architecture and the use of PROFIBUS and PROFINET networks to increase the efficiency and durability of the system, reducing costs and improving data transmission speed. In addition, the automation pyramid presented in the chapter underlines the importance of an efficient hierarchical integration of the different levels of an industrial system.

Chapter 4: Technical and economic analysis

4.1 Introduction

This chapter addresses the key aspects of the economic analysis of this research, focusing on the costs and impacts of the proposed solutions. A well-structured analysis provides financial tools to estimate the total cost associated with the creation of a project. In addition, it is essential for managing expenses efficiently, avoiding the acquisition of unnecessary materials and aligning with the guidelines of the country's economic policy.

4.2 Total cost

The total cost of a project represents the sum of all the economic resources needed to carry it out, including both direct and indirect expenses.

CT = CD + CI (ec. 4.1)

Where:

- CT: total cost
- CD: direct costs
- CI: indirect costs

CD = SB + SC + SS + MD + DP + OG (ec. 4.2)

Where:

- SB: basic salaries
- SC: complementary salaries (payment to the vacation fund)
- SS: social security
- MD: direct materials
- DP: per diems and travel expenses.
- OG: Other Expenses.

4.2.1 Calculating SB

The total basic salary (SB) represents the base remuneration assigned to the students for the time invested in the project, excluding additional benefits such as vacations or social security contributions. This value includes the sum of the base salaries of all the people who participate in the preparation of the project proposal.

SB = :2†“Ó âæâ’ ’ r &’ †V2à 4.3)

Where:

- Ai: days devoted to the research by participants.
- Bi: daily salary of the participant (monthly salary / 24).
- N: total number of participants.

Author Laura Díaz Pérez:

A1 = 56 days

B1 = 400 CUP / 24 = 16.667 CUP

Supervisor Carmenchu Regueiro Busoch:

A2 = 56 days

B2 = 6767.8 CUP / 24 = 281.991 CUP

Supervisor Romar Alexei Borges Rivero:

A3 = 56 days

B3 = 4500 CUP / 24 = 187.5 CUP

Supervisor Gonzalo de la Rosa Izquierdo:

A4 = 56 days

B4 = 5560 CUP / 24 = 231.667 CUP

Therefore:

SB = (56 days × 16.667 CUP) + (30 days × 281.991 CUP) + (30 days × 187.5 CUP) + (30 days × 231.667 CUP)

SB = 21968 CUP

4.2.2 Calculating SC

The complementary salary (SC) equals 9.09% of the basic salary and is intended to cover vacation pay, according to the standards and regulations established at the national level.

SC = 0.0909 × SB (ec. 4.4)

SC = 0.0909 × 21968 CUP

SC = 1996.899 CUP

4.2.3 Calculating SS

Social security equals 10% of the basic salary including the complementary salary.

SS = 0.05 × (SB + SC) (ec. 4.5)

SS = 0.05 × (21968 CUP + 1996.899 CUP)

SS = 1198.245 CUP

4.2.4 Calculating MD

Expenses on direct materials (MD) correspond to the financial resources allocated to the purchase of equipment, components, and technical tools that are used directly in carrying out the project. In this case, it was necessary to replace two GE Fanuc PLCs and their modules with the Siemens equivalents in the S7-1500 line. The purchase of accessories for the correct functioning of the communications was also necessary, and for the client-server architecture proposal the purchase of an HP PC was estimated. Table 4.1 lists these devices, together with their individual prices and the total cost according to the required quantity of each one.

Propuestas	Modelo/Referencia	Cantidad	Precio por unidad/ met	Precio total
CPU 1516-3 PN/DP	6ES7516-3AP03-0AB0	1	$ 3,161.20	$ 3,161.20
Memory Card para CPU	6ES7954-8LC03-0AA0	1	$ 79.75	$ 79.75
DI 16x24VDC	6ES7521-1BL00-0AB0	16	$ 283.14	$ 4,530.24
DQ 16x24VDC/0.5A	6ES7522-1BH00-0AB0	16	$ 223.00	$ 3,568.00
TM PosInput 2	6ES7550-1AA00-0AB0	7	$ 562.48	$ 3,937.36
IM 155-6 PN ST	6ES7155-6AU00-0CN0	2	$ 378.31	$ 756.62
PS 60W 24VDC	6ES7505-ORA00-0AB0	1	$ 206.78	$ 206.78
DI 32x24VDC BA	6ES7521-1BL10-0AA0	3	$ 284.26	$ 852.78
DQ 32x24VDC/0.5A BA	6ES7522-1BL10-0AA0	4	$ 215.00	$ 860.00
Front connectors 40 pole	6ES7592-1AM00-0XB0	39	$ 63.00	$ 2,457.00
Riel DIN 1.5 m	6ES7590-1AB60-0AA0	3	$ 45.33	$ 135.99
Switch PROFINET	SCALANCE XB005	1	$ 107.35	$ 107.35
Terminales PROFINET	6GK1901-1BB10-2AA0	10	$ 38.93	$ 389.30
Terminales PROFIBUS	6ES7972-0BA52-OXA0	15	$ 21.25	$ 318.75
Licencia WinCC	-	1	$ 258.67	$ 258.67
Licencia TIA Portal	STEP 7 Professional	1	$ 353.40	$ 353.40
Cable Profinet	-	100 m	$ 5.00	$ 500.00
Cable Profibus	-	60 m	$ 7.00	$ 420.00
PROFIBUS-DP Gefran	TPD32-PB	4		$ 4.00
PROFIBUS-DP ABB RDCO-02	RDCO-02	6		$ 6.00
PROFIBUS-DPABB NNPB-02	NNPB-02 HP EliteDesk 8	2		$ 2.00
HP EliteDesk 800 G6 Tower	-	1	$ 1,169.00	$ 1,169.00
Precio Total			$	24,074.19

Table 4.1 Material costs.

Therefore:

MD = $24074.19

4.2.5 Calculating DP

Per diems and travel (DP) include the expenses related to special per diems and worker transportation. In this project it is not necessary to make investments in this aspect, so no costs associated with per diems and travel are generated.

DP = 0

4.2.6 Calculating OG

Other expenses (OG) covers the cost of equipment use, specifically the expense associated with machine time, which has a value of 10.00 CUP for each working hour. However, in this case this type of expense is not incurred, so the amount is zero.

OG = 0

4.2.7 Calculating CD

CD = 21968 CUP + 1996.899 CUP + 1198.245 CUP + $24074.19 + 0 + 0

CD = 25163.144 CUP + $24074.19

4.2.8 Calculating CI

This refers to the general expenses indispensable for the execution of the project, but which are not directly linked to a particular activity, such as electricity consumption, supervision costs, or the use of infrastructure. These expenses are calculated by applying a coefficient to the base salary of the research project.

CI = 0.84 × SB (ec. 4.5)

CI = 0.84 × 21968 CUP

CI = 18453.12 CUP

4.2.9 Calculating CT

CT = 25163.144 CUP + $24074.19 + 18453.12 CUP

CT = 49237.334 CUP + $24074.19

4.3 Project feasibility

4.3.1 Economic and financial feasibility

The project, although it implies a considerable cost, is profitable, since the entity has the financial capacity to cover the total investment required.

4.3.2 Technical feasibility

The "José Martí" Steel Company, also known as Antillana de Acero, has the technical and human resources necessary to carry out the project effectively.

4.3.3 Environmental feasibility

The project is designed to comply with the established regulations and, with adequate execution, poses no risk to the environment.

4.3.4 Political feasibility

The project is aligned with the objectives and policies of the organization, without compromising its image or institutional prestige.

4.3.5 Legal feasibility

The implementation of the project does not imply changes or adjustments that may affect compliance with the legal regulations in force at the organization.

4.4 Project impact

4.4.1 Economic impact

The modernization of the rolling mill at the Antillana de Acero Company will have a transcendental economic impact, since it increases the efficiency and productive capacity of the company, reducing operating costs and improving product quality. This enables the company to compete in more demanding markets, increasing its revenue and strengthening its position in the steel industry. In addition, investment in modern technology and equipment generates a positive effect on the supply chain, benefiting suppliers and customers. At the macroeconomic level, the project contributes to the growth of the industrial sector, generating direct and indirect jobs and promoting the economic development of the region. Altogether, this modernization not only boosts the company's profitability, but also strengthens the local and national economy.

4.4.2 Social impact

The modernization of the rolling mill at the Antillana de Acero Company will have a significant social impact, since it not only improves the working conditions of the workers, but also promotes training in new technologies. By implementing more efficient and safer processes, a healthier and more productive work environment is promoted, which directly benefits

employees and their families. In addition, by strengthening the competitiveness of the company, the economic development of the community is contributed to, creating a multiplier effect in the region. This project reinforces the company's commitment to social well-being and people's progress, consolidating its role as a key actor in local development.

4.4.3 Environmental impact

The modernization of the rolling mill at the Antillana de Acero Company will represent an opportunity to reduce the environmental impact of its operations. By incorporating more efficient and sustainable technologies, energy consumption will be decreased, contributing to a cleaner production process. In addition, the project includes compliance with environmental regulations, ensuring that the exploitation of resources is carried out responsibly. With these improvements, the company will not only optimize its production, but will also reinforce its commitment to environmental protection and sustainable development.

4.5 Partial conclusions

The economic study carried out in this project demonstrated its viability, concluding that the company will obtain benefits if it decides to implement it, since it will ensure greater stability in the production of steel profiles in the rolling mill. This advancement will not only drive an increase in the company's income, but will also have a positive impact on the country's economy.

General conclusions

In this work, a study was carried out on the automation and electrical drive systems applied to the process. The knowledge gathered during the research served as the basis for the design of a proposal for an automation and electrical drive system in the subprocesses prior to line 320 of the 300 rolling mill at Antillana de Acero. From this analysis it can be concluded that, once the technical automation means, the control strategy, and the monitoring system have been selected, it is feasible to design an automation and electrical drive system for the auxiliary equipment of line 320 of the 300 rolling mill at Antillana de Acero. The proposed system outperforms the current one, since replacing the existing controllers with a more modern one and implementing more advanced monitoring software solves problems such as frequent failures, lack of spare parts, the limitations of the SNP-X protocol, and the restrictions of the software used to program the PLCs. In addition, the proposal to include a second PC as a server guarantees communication superior to the one existing in the current system, considering that a communication network was also designed to integrate the subprocesses into the monitoring level within the automation pyramid. The proposed system outperforms the current one from the energetic and ecological point of view, optimizing the use of resources and reducing the environmental impact. Finally, the project is economically viable, which reinforces its feasibility of implementation.

Recommendations

To extend the results obtained through the study carried out and to optimize their use, it is recommended:

- To carry out the corresponding programming of the PLCs to be used, following the line of the flowcharts presented and implementing the algorithms of those equipments not covered in this work.

- To implement, either in the existing monitoring software or in a completely new one, a fault detection system and even a self-regulating system, which would mean a great technological advancement for the rolling mill and the company in general.

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Annexes

Annex 1: Nameplate data of the Elektromaschinenbau motors.

Annex 2: Table with the power rating of the Gefran converters.

Annex 3: PROFINET interface module IM 155-6PN, SIMATIC ET 200SP.

Annex 4: HP EliteDesk 800 G6 Tower PC product sheet.

[The annexes contain tables, photographs, and diagrams that are not reproduced in this unofficial translation; please refer to the original Spanish version.]